



Koneru Lakshmaiah Education Foundation

(Deemed to be University estd. u/s. 3 of the UGC Act, 1956)

Off-Campus: Bachupally-Gandimaisamma Road, Bowrampet, Hyderabad, Telangana - 500 043.

Phone No: 7815926816, www.klh.edu.in

DEPARTMENT OF CSE

PROJECT ABSTRACT SUBMISSION

COURSE: Data Sciences (24AD3107)

A.Y. 2026 - 2027

BATCH: 8

TITLE OF THE PROJECT		Student Exam Score Predictor	
S. No.	University ID	Name	
1	2420030053	K. Ysaswi	
2	2420030037	G. Poojitha	
3	2420030499	S. Anjana	
4			
Name of the Guide		Dr. N. Sunanda	

Abstract

A student's academic performance is influenced by several factors, including how consistently they study, how regularly they attend classes, and their broader personal and demographic background. However, these factors are rarely analyzed together in a structured way that allows performance to be predicted in advance. This project aims to build a regression-based system that estimates a student's expected exam score using such academic and demographic information.

The process begins with collecting a dataset containing details like daily study hours, attendance percentage, gender, parental background, and past academic performance. This raw data is cleaned to handle missing or inconsistent values, and categorical attributes are converted into a numerical form suitable for machine learning. Exploratory data analysis is then carried out to understand how each feature relates to the final exam score, helping identify which factors have the strongest influence on performance. Once the data is prepared, regression algorithms such as Linear Regression and Random Forest Regression are applied to learn the underlying relationship between the input features and exam scores. Multiple models are trained and compared so that the most accurate and reliable one can be selected. The performance of each model is evaluated using standard metrics such as Mean Squared Error (MSE) and R-squared (R^2) score, which indicate how closely the predicted scores match the actual results.

The final system is designed to take a student's study and attendance details as input and return a predicted exam score as output. Such a tool can help students understand how their habits affect their performance and encourage them to make positive changes early on. It can also assist educators in identifying students who may be at risk of underperforming, allowing timely academic support and intervention. Overall, this project demonstrates how a simple regression model can turn everyday academic data into meaningful, actionable insights for both students and teachers.